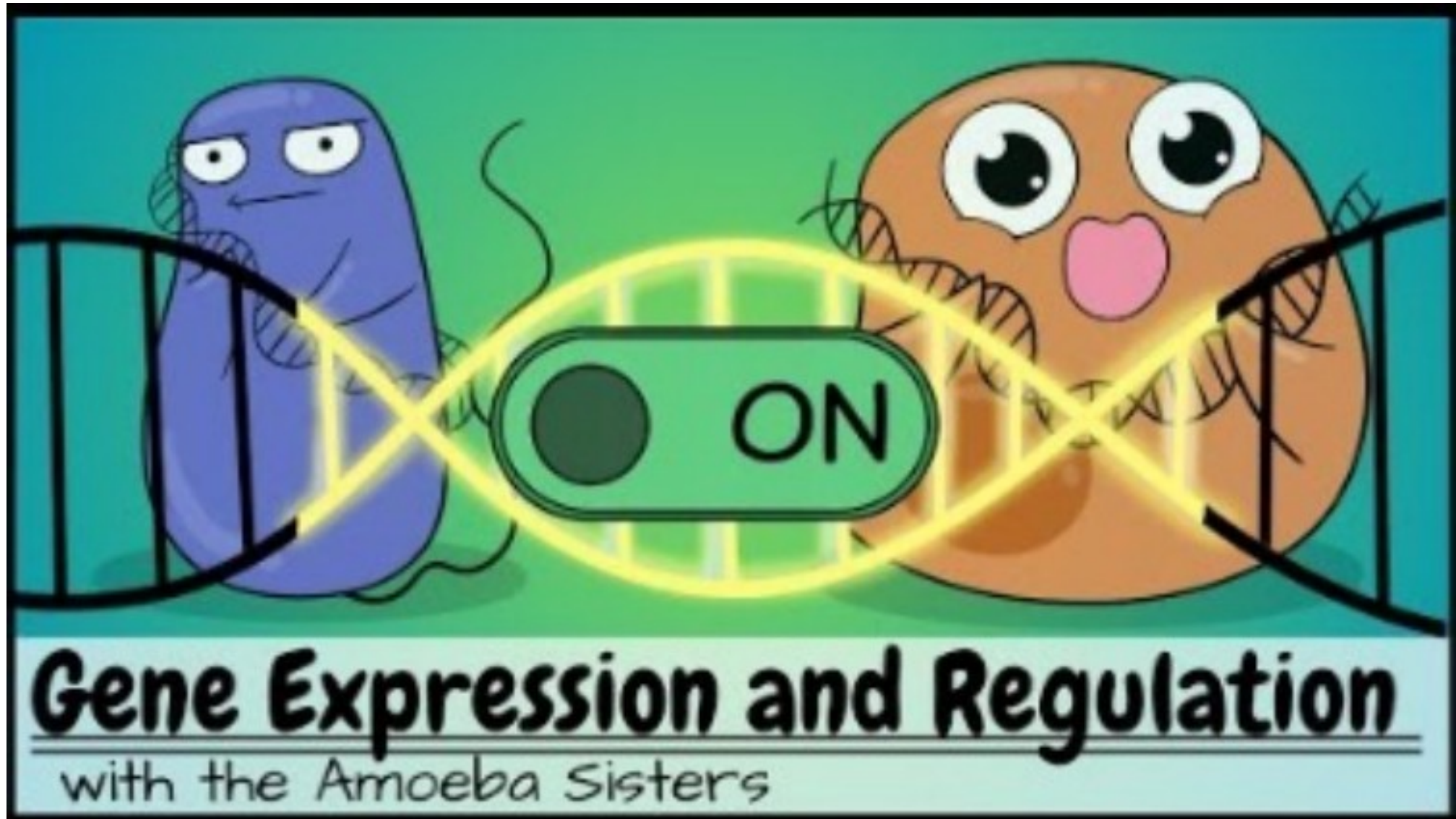
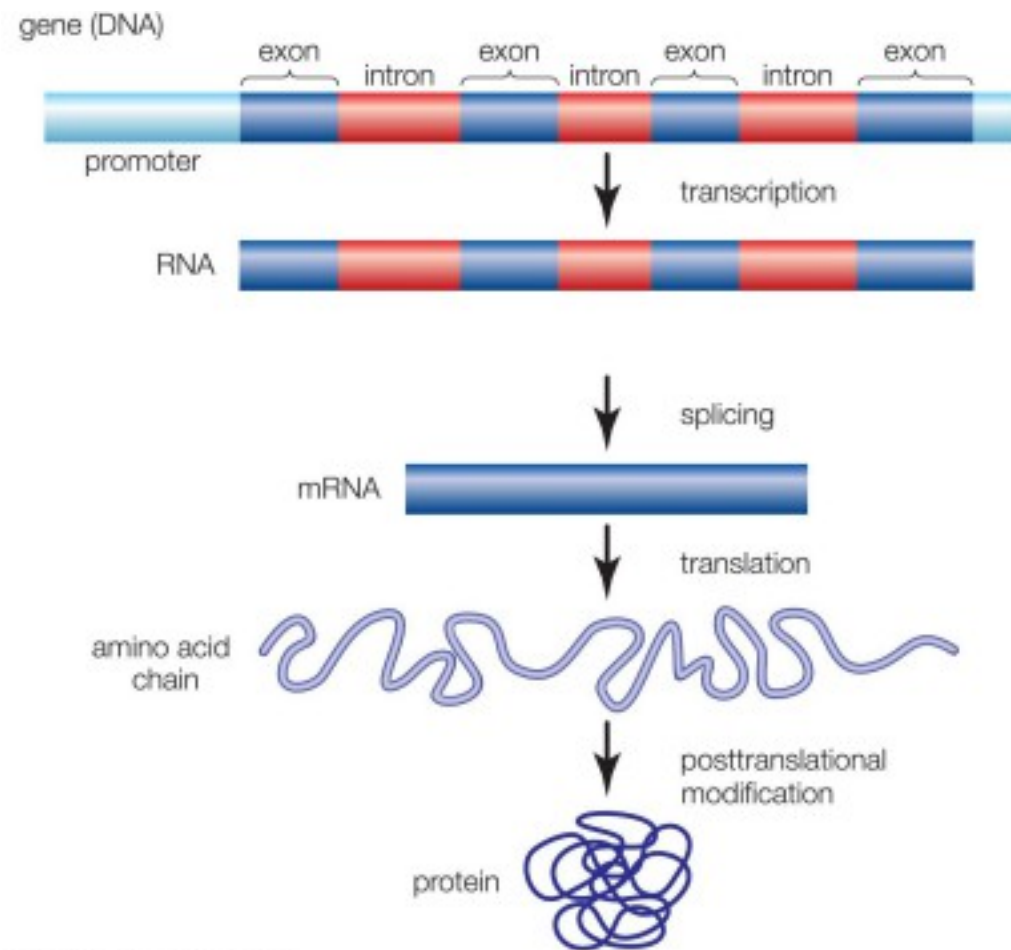
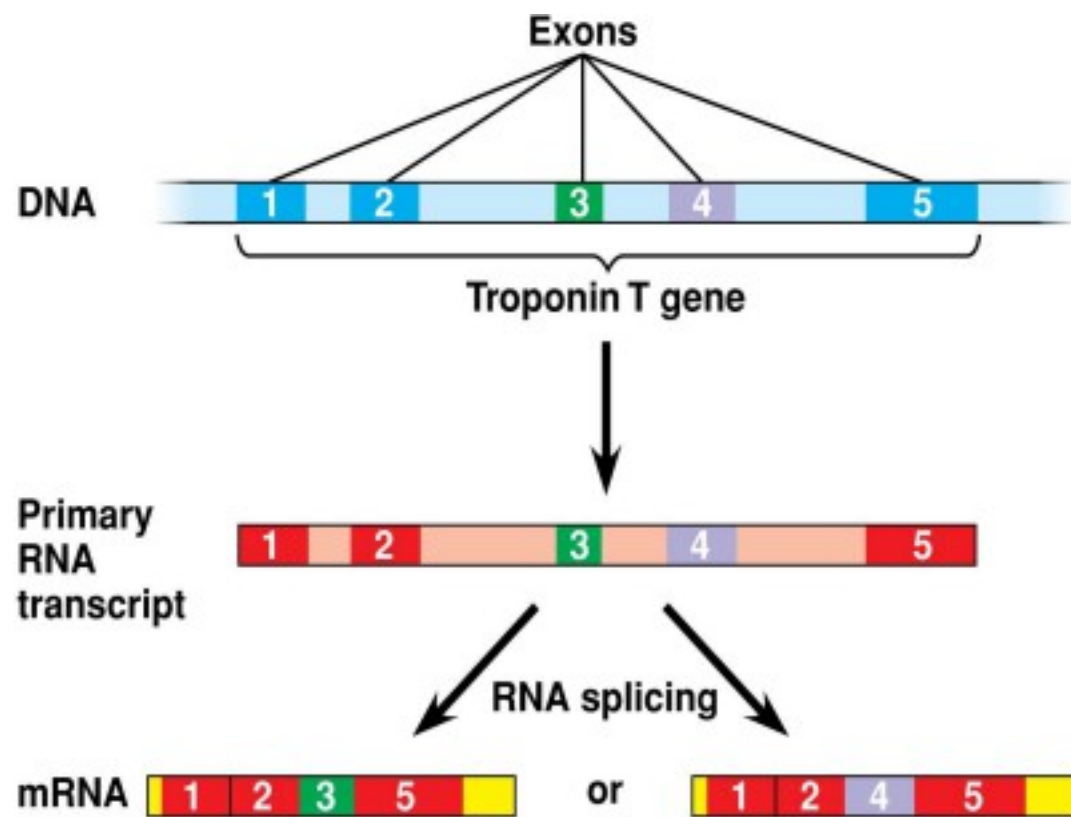




<https://www.youtube.com/watch?v=ebIpkw3Xap>



Control in living systems

Gene expression

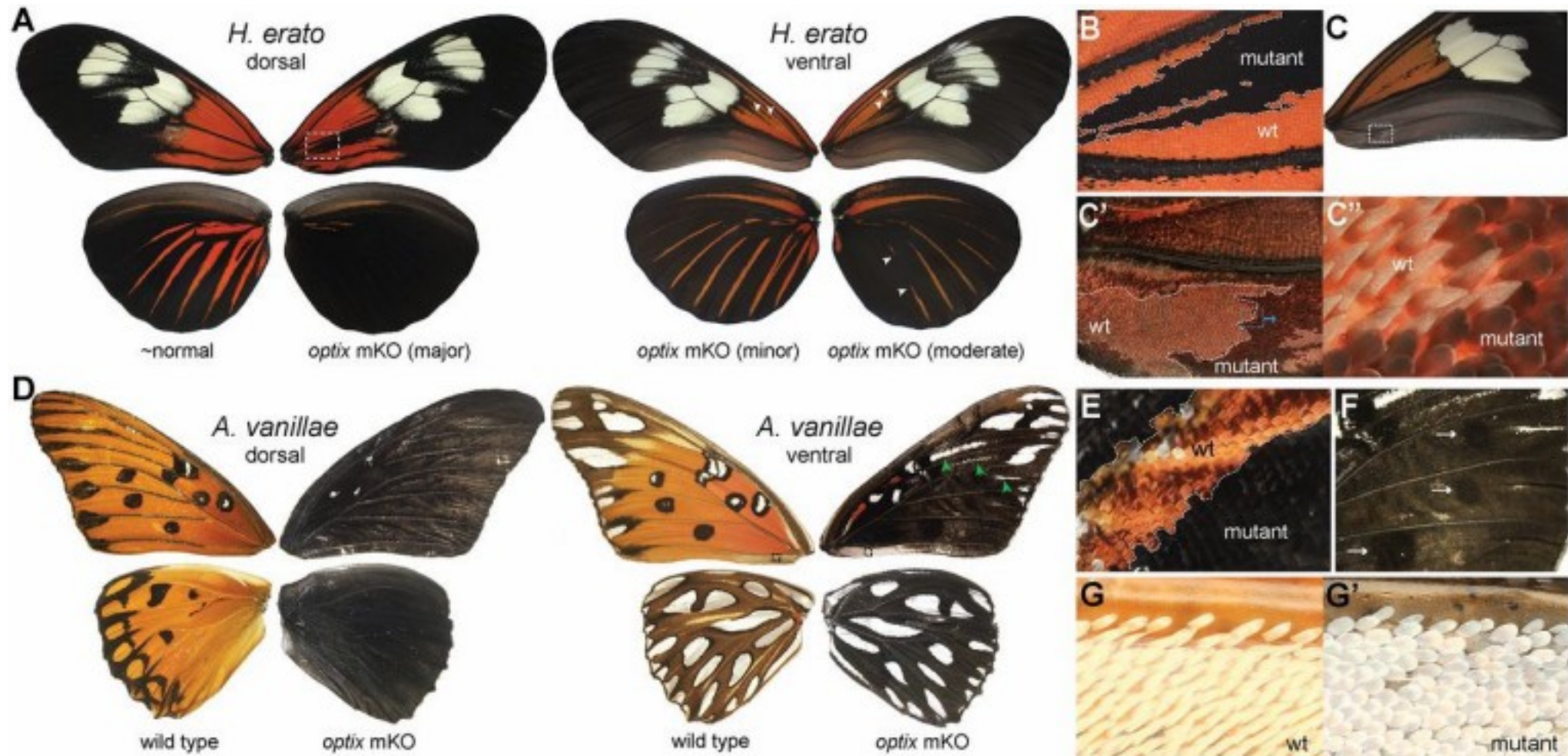
control of cell division

Hormones, signals & receptors

feedback control

control failure & disease





<https://doi.org/10.1073/pnas.170905811>

Gene expression

Information encoded in a gene is turned into a function.

on/off switch” to control when and where RNA molecules and proteins are made and as a “volume control” to determine how much of those products are made.

carefully regulated, changing substantially under different conditions and cell types.

The RNA and protein products of many genes serve to regulate the expression of other genes.

Where, when, and how much a gene is expressed can also be assessed by measuring the functional activity of a gene product or observing a phenotype associated with a gene.

The DNA strand that corresponds to the mRNA is called the coding or sense strand.

Gene expression

Operon – grouped genes that are transcribed together – code for functionally similar proteins

Promoter – section of DNA where RNA polymerase binds

Operator – Controls activation of transcription

- on/off switch
- between promoter and genes for proteins – structural genes

Repressor protein – binds to operator to block RNA polymerase and shut down transcription

- Turns off the operon

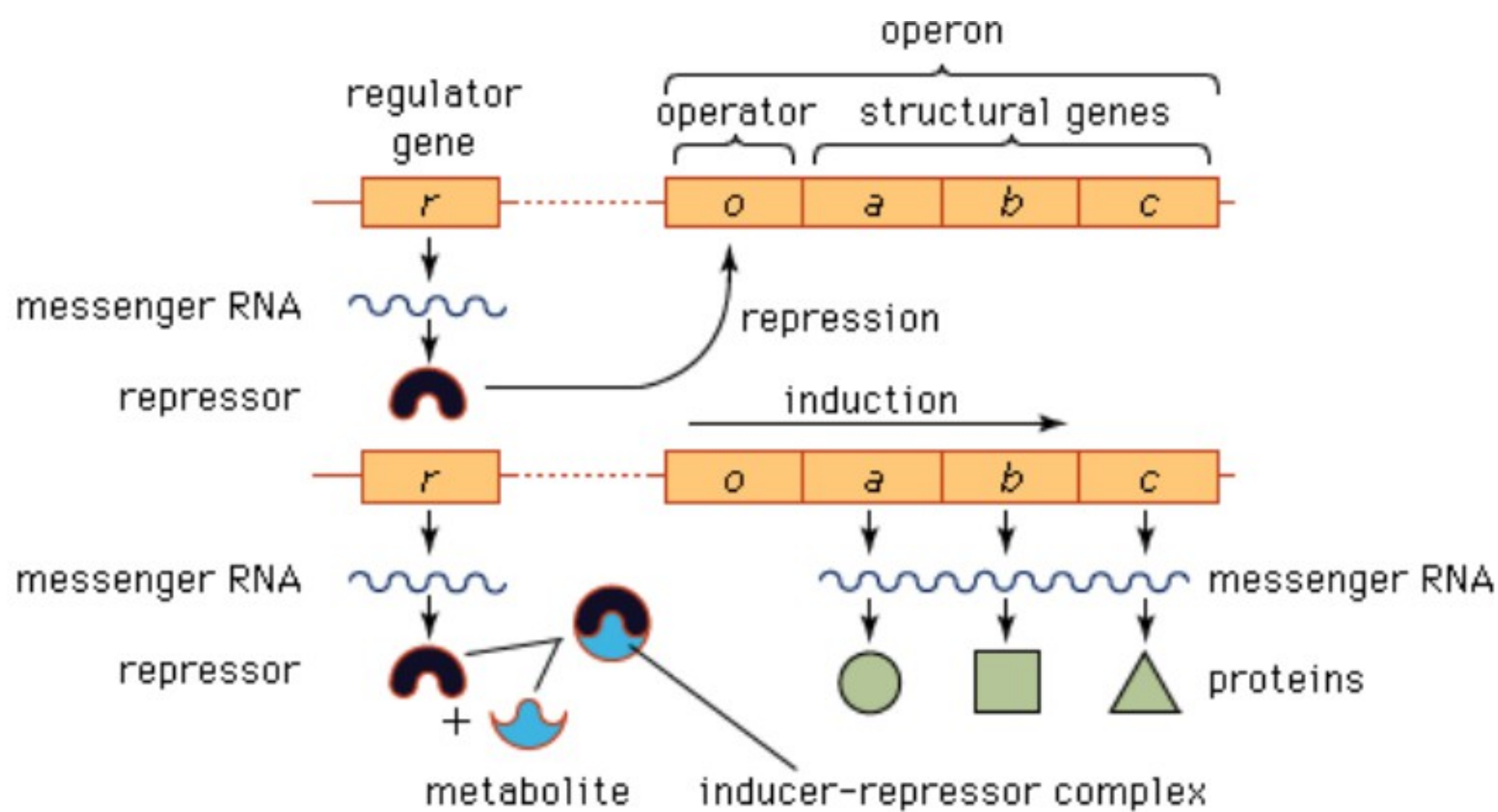
Corepressor – keeps the repressor protein on the operator (Trp operon)

Inducer – pulls repressor off the operator

- Turns on the operon – lactose on the lac operon

Regulatory gene – produces the repressor protein

Structural genes – code for proteins



Lac Operon

Gene z: Galactosidase

Lactose- Glucose

Gene y: Permease

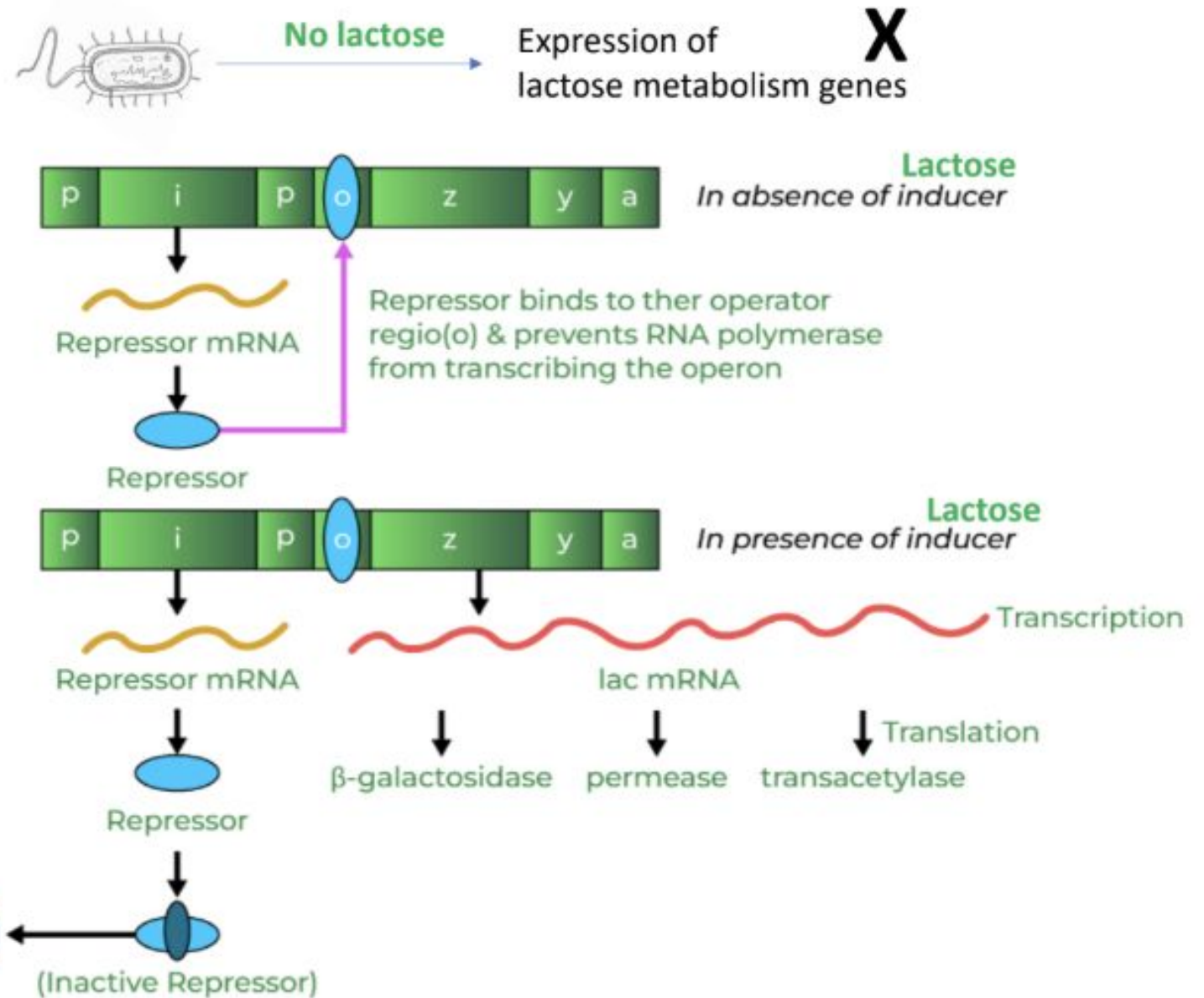
Lactose crossing membrane

Gene a: Transacetylase

Helps Galactosidase



Lactose



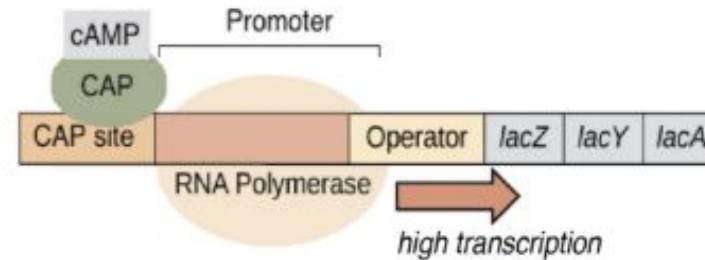
Does the gene expression occur at a constant rate?

No!

- **Catabolite activator protein (CAP)** is a positive regulator
- Low glucose (Hunger Signal) cAMP enables CAP and DNA binding
- CAP increases the rate of gene expression

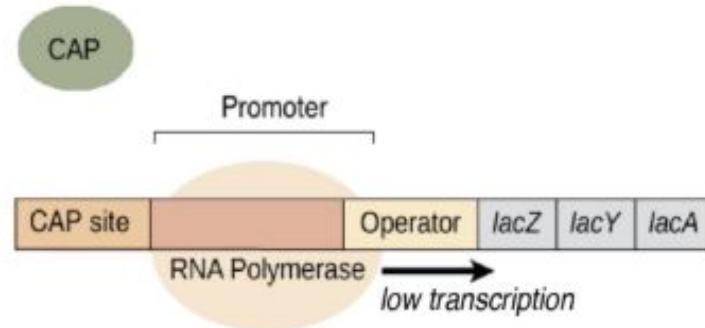
Low glucose:

When glucose levels are low, cAMP is produced. The cAMP attaches to CAP, allowing it to bind DNA. CAP helps RNA polymerase bind to the promoter, resulting in high levels of transcription.



High glucose:

When glucose levels are high, no cAMP is made. CAP cannot bind DNA without cAMP, so transcription occurs only at a low level.



Does all the DNA gets transcribed?



No!

Only certain sections of the DNA are made (transcribed) into message (mRNA)

AND...only part of the mRNA is actually used and sent out of the nucleus to meet up with a ribosome!

This is EDITING!!

How is mRNA Edited?

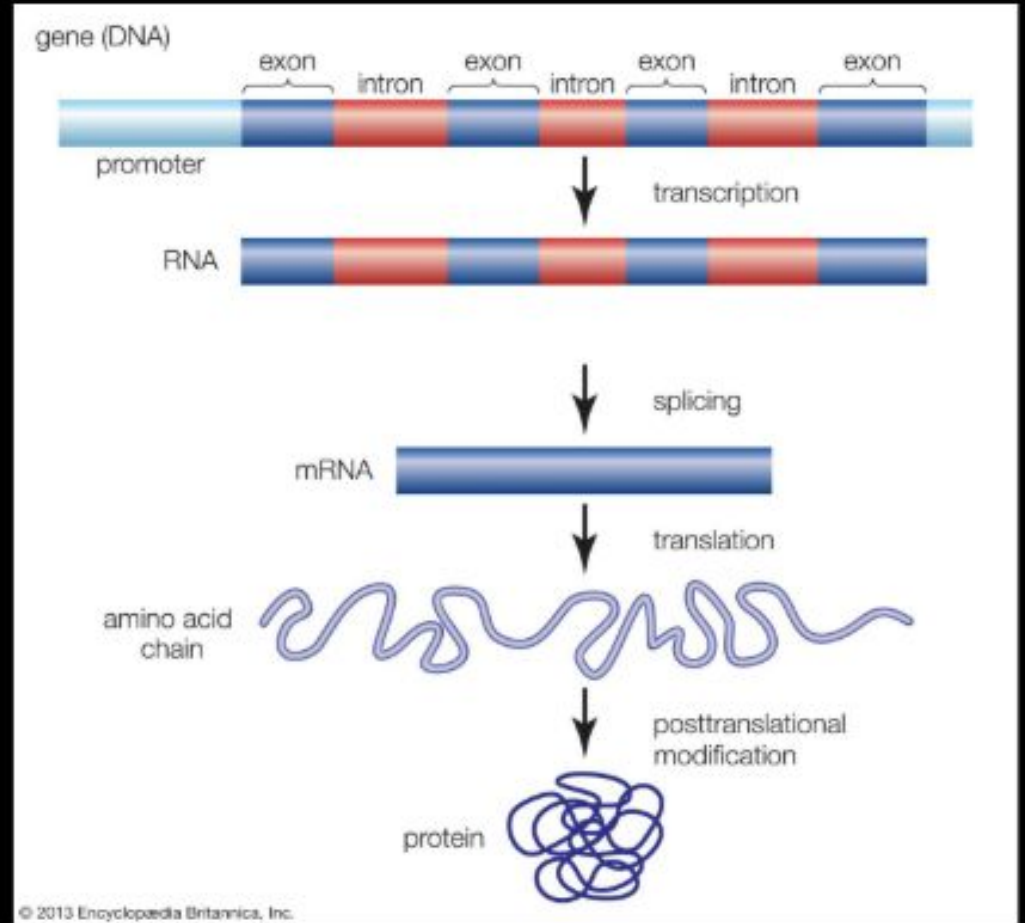
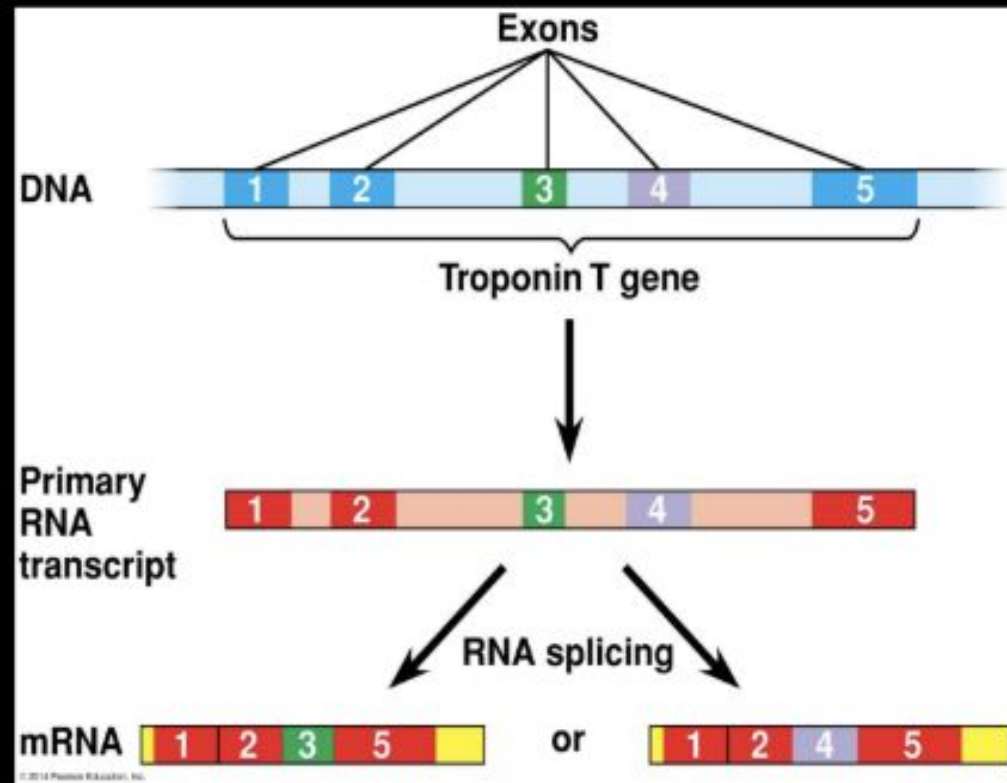
On a mRNA strand there are areas called: Exons and Introns

Introns are cut out before leaving the nucleus

Exons are left, and this shortened piece of mRNA leaves the nucleus and gets
Translated into Proteins

RNA Processing Regulation

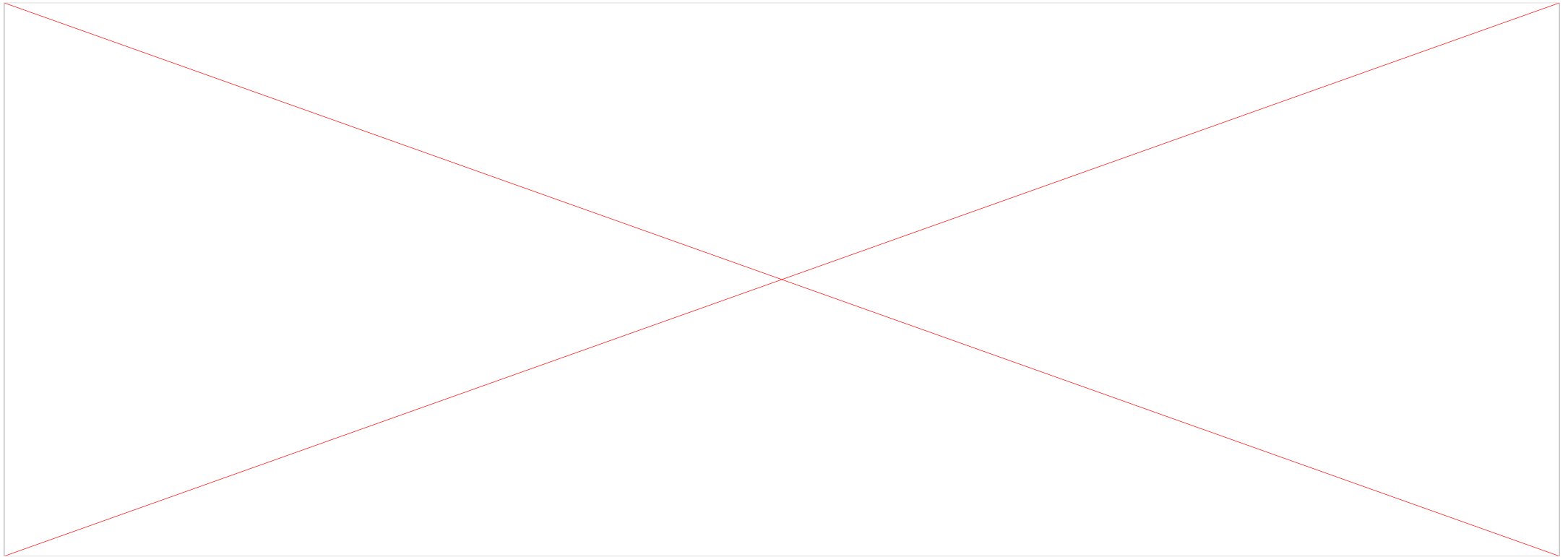
Alternative RNA Splicing – different regions of the pre-mRNA serve as introns or exons creating different mRNA strands depending on what is removed & spliced together



Gene expression logic gates



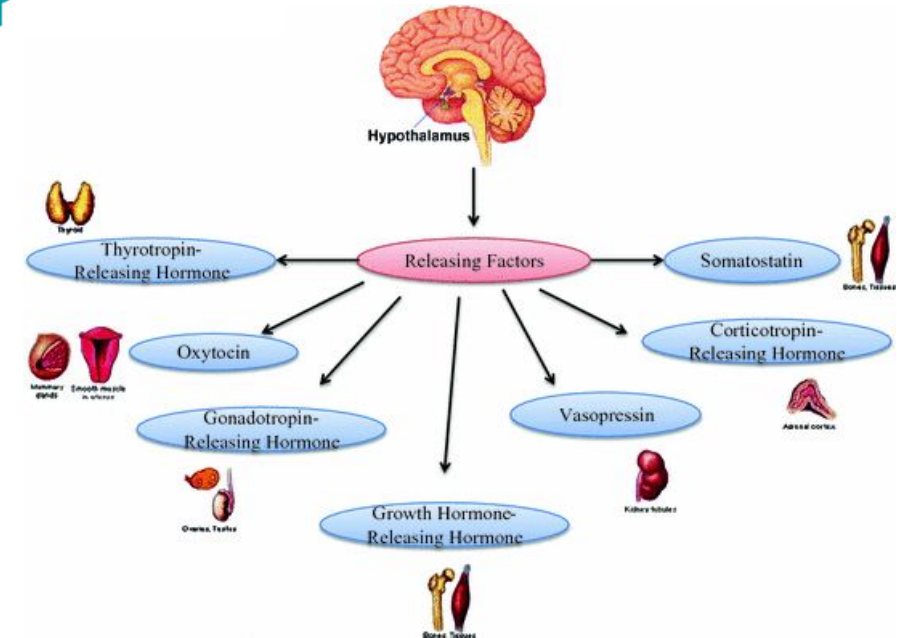
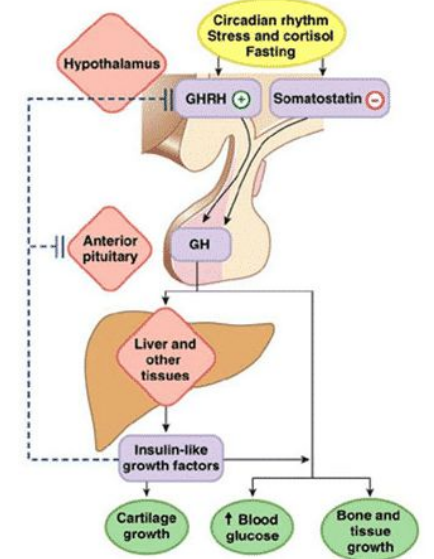
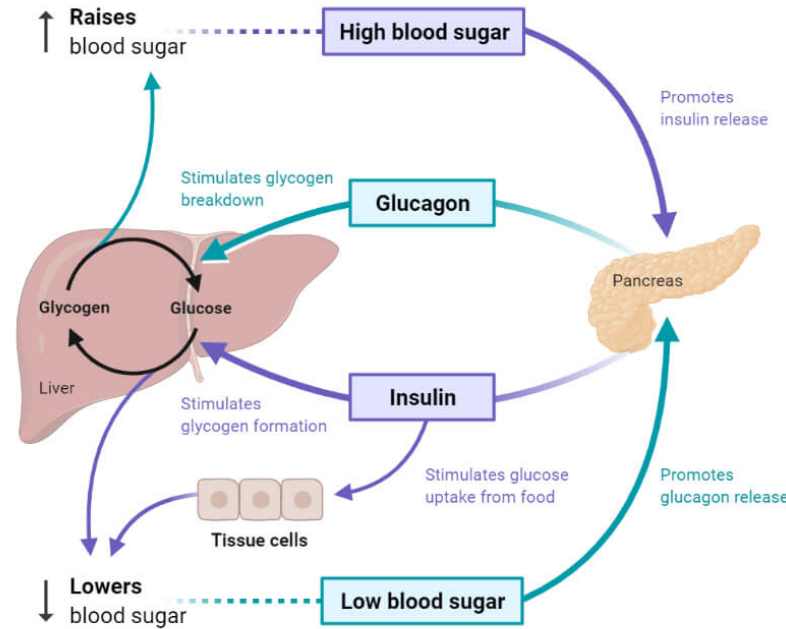
- Engineered genetic circuits that mimic the functionality of traditional electronic logic gates using biological components.
- Built using well-characterized genetic elements and regulatory proteins.



Hormones

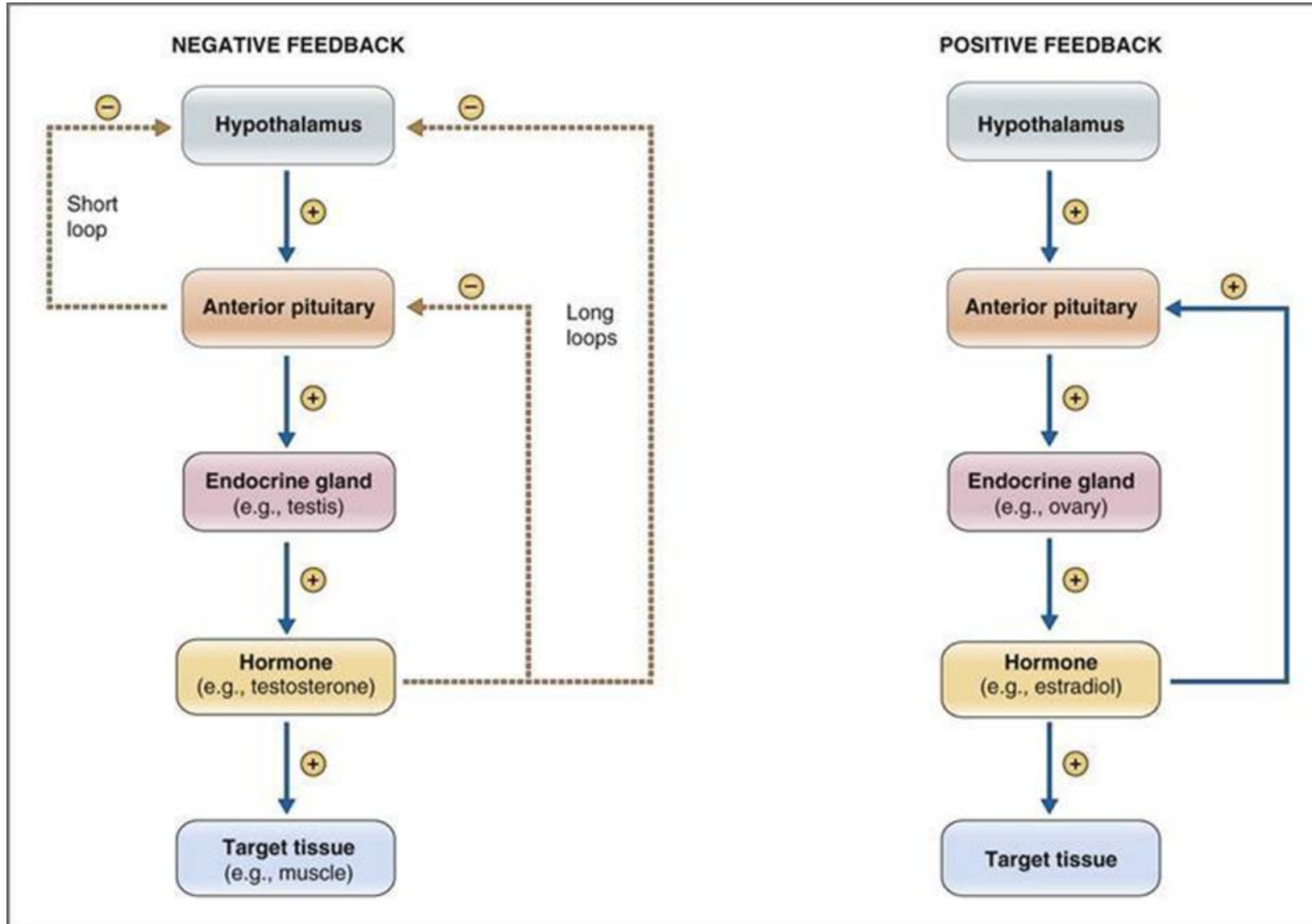
- Signaling molecules produced by glands in the endocrine system.
- Regulate various physiological processes in organisms.
- Travel through the bloodstream to target cells, where they exert specific effects.
- Play a crucial role in maintaining homeostasis by coordinating and integrating the activities of different organs and tissues.
- Hormones bind to specific receptors on target cells.
- Hormone regulation often involves feedback mechanisms.
- Hormones can be classified into two main types: steroid hormones (derived from cholesterol) and peptide hormones (made up of amino acids).

Regulation of Blood Glucose

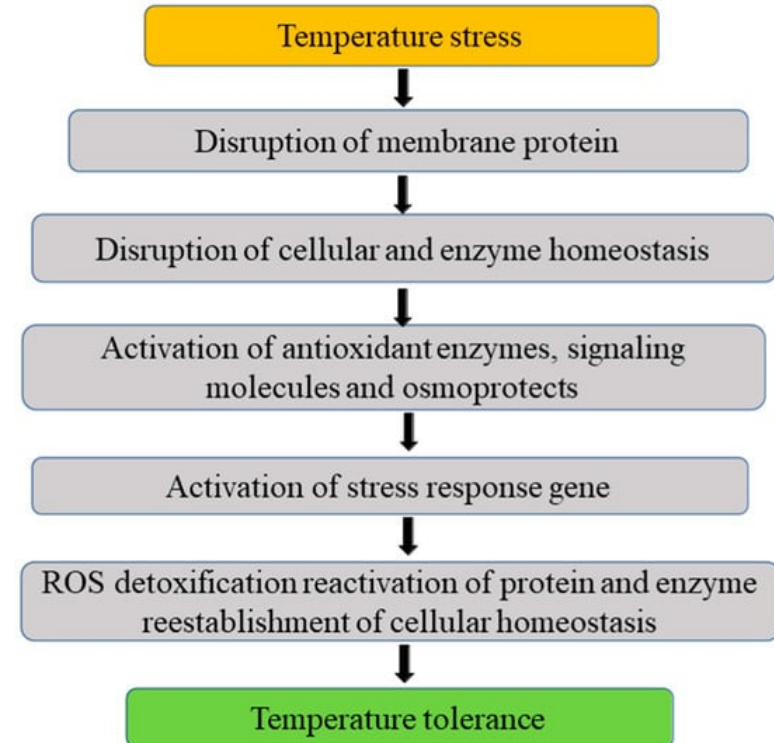
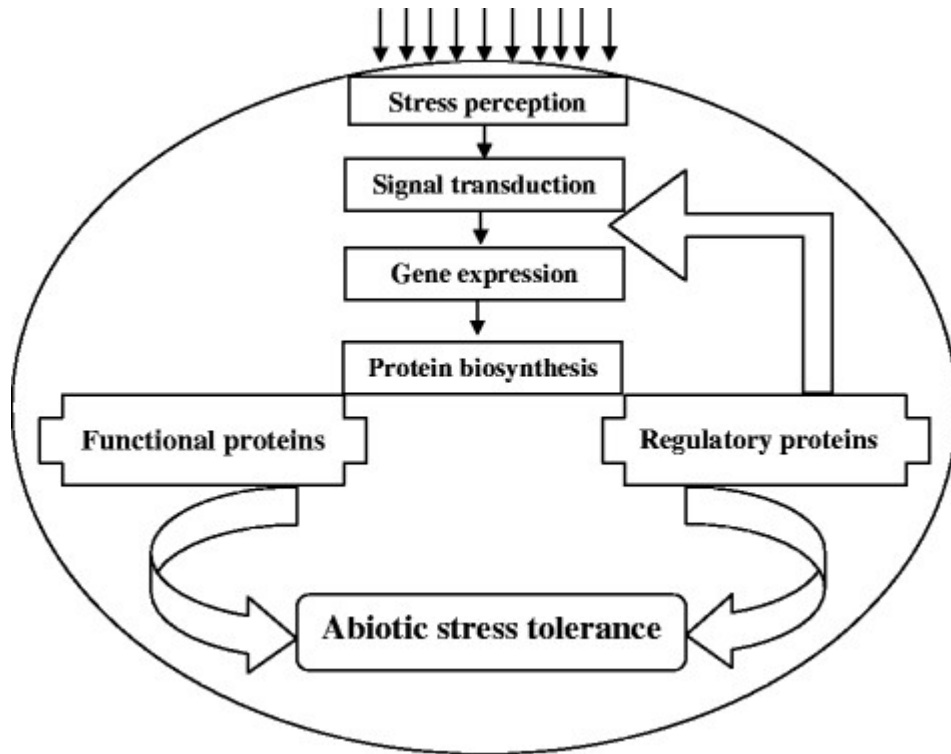


Feedback control

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Stress response, tolerance and adaptation



- The **stress response**, often referred to as the "fight-or-flight" response, is a physiological and psychological reaction that occurs in response to a perceived threat or stressor. This response is designed to help an organism cope with challenging or threatening situations by preparing the body for action.
- **Tolerance** refers to the reduced sensitivity or diminished response to a substance or stimulus over time.
- **Adaptation** refers to the process by which organisms evolve and develop traits that improve their survival and reproductive success in a given environment.

Rhythms and cycles



- Organisms exhibit various rhythms and cycles in their biological processes. These patterns are often influenced by environmental factors, internal regulatory mechanisms, and evolutionary adaptations.
- Circadian rhythms** are approximately 24-hour cycles that regulate various physiological and behavioral processes, such as sleep-wake cycles, hormone secretion, and metabolism.
- Ultradian rhythms** are shorter cycles that occur more than once within a 24-hour period. Examples include the cycles of feeding, hormone secretion
- Infradian rhythms** have a period longer than 24 hours, with cycles occurring less frequently than once per day. Examples include the menstrual cycle in females, seasonal reproductive cycles, and hibernation cycles in certain animals.
- Circannual rhythms** have a yearly cycle and are often associated with seasonal changes. Examples include the migration of birds, the shedding of fur or feathers

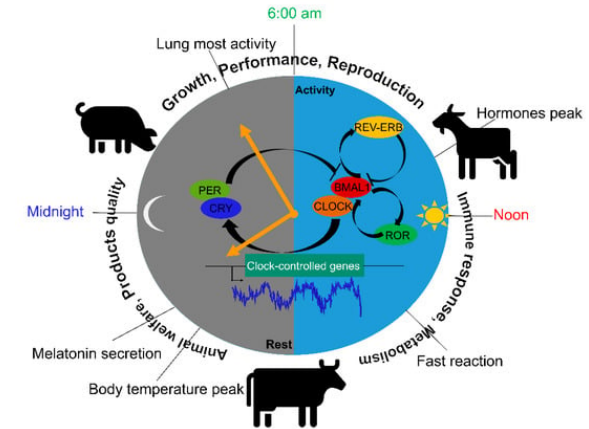


FIGURE 2
THE ULTRADIAN PERFORMANCE RHYTHM

